

ELEC 221 Tutorial 2 – Solutions

Linearity, Time Invariance, Impulse Response, and Convolution

Monday, 21 September 2026

Overview

1. Linearity and time invariance
2. Impulse, step, and signal manipulation
3. The convolution sum

1.1 – Linear? Time invariant?

(a) $y(t) = t^2 x(t - 1)$

- Linear: $ax_1 + bx_2 \rightarrow t^2[ax_1(t - 1) + bx_2(t - 1)] = ay_1 + by_2 \checkmark$
- Time invariance: input $x(t - t_0)$ gives $t^2 x(t - t_0 - 1)$, but $y(t - t_0) = (t - t_0)^2 x(t - t_0 - 1)$. These differ because of the explicit $t^2 \Rightarrow$ **linear, not TI**.

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(b) $y[n] = x[n + 1] - x[n - 1]$

- A weighted sum of shifted inputs $\Rightarrow$ linear.
- Input $x[n - n_0]$ gives $x[n - n_0 + 1] - x[n - n_0 - 1] = y[n - n_0] \Rightarrow$ TI.
- **LTI**.

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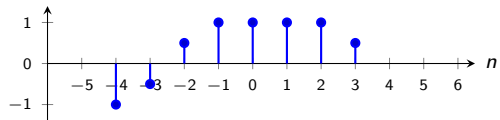
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(c) $y(t) = \sin(x(t))$

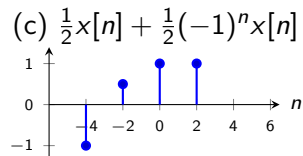
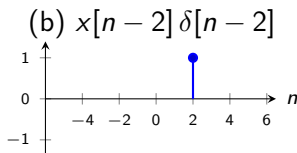
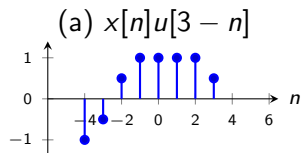
- Not linear: with $x \equiv \pi/2$, $y \equiv 1$, but input $2x \equiv \pi$ gives $\sin \pi = 0 \neq 2$.
- TI: $\sin(x(t - t_0)) = y(t - t_0) \Rightarrow$ **TI, not linear**.

2.1 – The signal $x[n]$ (read off Figure 1)



$$x[n] = \begin{cases} -1, & n = -4 \\ -\frac{1}{2}, & n = -3 \\ \frac{1}{2}, & n = -2 \\ 1, & -1 \leq n \leq 2 \\ \frac{1}{2}, & n = 3 \\ 0, & \text{otherwise} \end{cases}$$

2.1 – Sketches



(a) $u[3-n] = 1$ for $n \leq 3$, and $x[n] = 0$ for $n > 3$, so the product is just $x[n]$.

(b) Sifting: $x[n-2]\delta[n-2] = x[0]\delta[n-2] = \delta[n-2]$.

(c) $\frac{1}{2}(1 + (-1)^n)$ is 1 for even n , 0 for odd n : keep the even-indexed samples of $x[n]$.

3.1 – $y[n] = x[n]x[n-2]$

(a) Review

- Memoryless? **No**: depends on $x[n-2]$.
- Causal? **Yes**: only present and past samples.
- Stable? **Yes**: $|x| \leq B \Rightarrow |y[n]| \leq B^2$.
- Invertible? **No**: $x_1 \equiv 0$ and $x_2 = \delta[n]$ both give $y \equiv 0$, since $\delta[n]\delta[n-2] = 0$.

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(b) LTI? No.

- Not linear (quadratic): input $ax[n]$ gives $a^2y[n]$, not $ay[n]$.
- It *is* time invariant: $x[n - n_0] x[n - n_0 - 2] = y[n - n_0]$.

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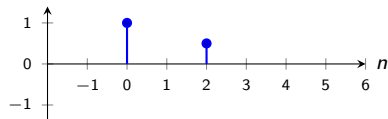
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(c) Input $A\delta[n]$: $y[n] = A^2\delta[n]\delta[n-2] = 0$ for all n .

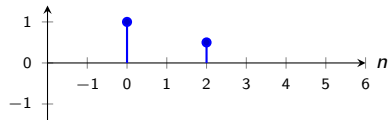
- The “impulse response” is identically zero, yet the input $\delta[n] + \delta[n-2]$ gives $y[n] = \delta[n-2] \neq 0$. Impulse responses only characterize *LTI* systems.

$$3.2 - h[n] = \delta[n] + \frac{1}{2}\delta[n - 2]$$



(a) A direct response followed by a delayed, weaker copy: an **echo**. Examples: room reverberation, multipath in wireless/radio, “ghosting” in analog TV, radar/sonar returns.

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(a) A direct response followed by a delayed, weaker copy: an **echo**. Examples: room reverberation, multipath in wireless/radio, “ghosting” in analog TV, radar/sonar returns.

(b) With $x[n] = (\frac{1}{2})^n u[n]$, $y[n] = x[n] + \frac{1}{2}x[n-2]$:

$$y[n] = \left(\frac{1}{2}\right)^n u[n] + \left(\frac{1}{2}\right)^{n-1} u[n-2] = \begin{cases} 1, & n = 0 \\ \frac{1}{2}, & n = 1 \\ 3\left(\frac{1}{2}\right)^n, & n \geq 2 \end{cases}$$

Check $n = 2$: $x[2] + \frac{1}{2}x[0] = \frac{1}{4} + \frac{1}{2} = \frac{3}{4} = 3\left(\frac{1}{2}\right)^2 \checkmark$

$$3.3 - h[n] = \left(\frac{1}{2}\right)^n u[n], x[n] = \delta[n] + \frac{1}{2}\delta[n-2]$$

Explicit sum: only $k = 0$ and $k = 2$ contribute:

$$y[n] = \sum_k x[k]h[n-k] = h[n] + \frac{1}{2}h[n-2] = \left(\frac{1}{2}\right)^n u[n] + \left(\frac{1}{2}\right)^{n-1} u[n-2].$$

Values: $y[0] = 1$, $y[1] = \frac{1}{2}$, $y[n] = 3\left(\frac{1}{2}\right)^n$ for $n \geq 2$.

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- **Check via properties:** this is exactly the result of 3.2(b) with the roles of x and h swapped. Commutativity, $x * h = h * x$, guarantees the same answer.
- **Also:** $h[n] * (\delta[n] + \frac{1}{2}\delta[n-2]) = h[n] + \frac{1}{2}h[n-2]$ by linearity and $h * \delta[n-k] = h[n-k]$.

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With a shifted x instead of a shifted h : $y[n] = \sum_k h[k]x[n-k]$, so

$$y[n] = \sum_{k=0}^{\infty} (\frac{1}{2})^k x[n-k] = \sum_{k=0}^{\infty} (\frac{1}{2})^k (\delta[n-k] + \frac{1}{2}\delta[n-2-k]),$$

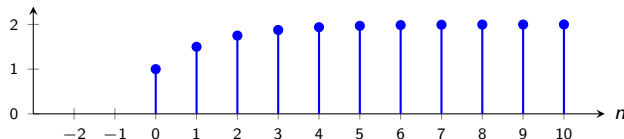
a weighted sum of delayed copies of x , which collapses to the same $y[n]$ (e.g.

$$y[3] = \frac{1}{8} + \frac{1}{2} \cdot \frac{1}{2} = \frac{3}{8} = 3(\frac{1}{2})^3 \checkmark).$$

$$3.4 - x[n] = \left(\frac{1}{2}\right)^{n-2} u[n-2], \quad h[n] = u[n+2]$$

$y[n] = \sum_{k=2}^{\infty} \left(\frac{1}{2}\right)^{k-2} u[n+2-k]$. The step is 1 iff $k \leq n+2$, so the sum runs over $2 \leq k \leq n+2$ (empty for $n < 0$). With $m = k - 2$:

$$y[n] = \sum_{m=0}^n \left(\frac{1}{2}\right)^m = \frac{1 - (1/2)^{n+1}}{1 - 1/2} = \left(2 - \left(\frac{1}{2}\right)^n\right) u[n].$$



Starts at $y[0] = 1$ and approaches 2 ($= \sum_{m \geq 0} \left(\frac{1}{2}\right)^m$) as $n \rightarrow \infty$.

3.5 – Setting up the sum: $x = 1$ on $[3, 8]$, $h = 1$ on $[4, 15]$

$$y[n] = \sum_{k=-\infty}^{\infty} x[k] h[n-k]$$

- $x[k] = 1$ exactly when $3 \leq k \leq 8$.
- $h[n-k] = 1$ exactly when $4 \leq n-k \leq 15$, i.e. $n-15 \leq k \leq n-4$.

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Every term in the sum is 0 or 1, so $y[n]$ just *counts* the integers k satisfying both conditions:

$$y[n] = \#\{k : \max(3, n-15) \leq k \leq \min(8, n-4)\} = \max\left(0, \min(8, n-4) - \max(3, n-15) + 1\right)$$

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Picture: the window $[n-15, n-4]$ (length 12) slides to the right as n increases, and $y[n]$ is its overlap with $[3, 8]$ (length 6).

- $\min(8, n-4)$ switches at $n-4 = 8 \iff n = 12$.
- $\max(3, n-15)$ switches at $n-15 = 3 \iff n = 18$.

3.5 – Casework on n

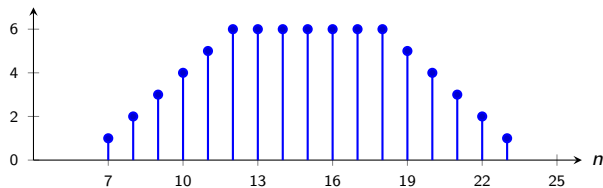
range of n	upper = $\min(8, n - 4)$	lower = $\max(3, n - 15)$	$y[n]$
$n \leq 6$	$n - 4 (\leq 2)$	3	upper < lower $\Rightarrow 0$
$7 \leq n \leq 12$	$n - 4$	3	$(n - 4) - 3 + 1 = n - 6$
$12 \leq n \leq 18$	8	3	$8 - 3 + 1 = 6$
$18 \leq n \leq 23$	8	$n - 15$	$8 - (n - 15) + 1 = 24 - n$
$n \geq 24$	8	$n - 15 (\geq 9)$	lower > upper $\Rightarrow 0$

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- Overlap starts when the window's right edge reaches the pulse's left edge: $n - 4 = 3 \Rightarrow n = 7$ ($= 3 + 4$).
- Overlap ends when the window's left edge passes the pulse's right edge: $n - 15 = 8 \Rightarrow n = 23$ ($= 8 + 15$) is the last nonzero sample.
- At the boundaries the neighbouring formulas agree: $n = 12$: $n - 6 = 6$; $n = 18$: $24 - n = 6$.

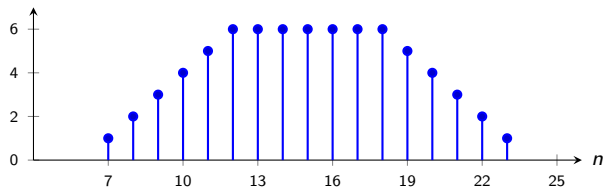
3.5 – Result



$$y[n] = \begin{cases} n - 6, & 7 \leq n \leq 11 \\ 6, & 12 \leq n \leq 18 \\ 24 - n, & 19 \leq n \leq 23 \\ 0, & \text{otherwise} \end{cases}$$

A trapezoid: ramps up, plateaus at the height 6 (the shorter pulse's length), then ramps down.

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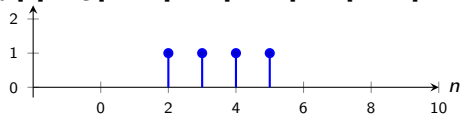
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- **Support check:** 7 to 23, i.e. $6 + 12 - 1 = 17$ samples.
- **Area check:** $\sum_n y[n] = (\sum x)(\sum h) = 6 \cdot 12 = 72$, and $(1+2+3+4+5) + 7 \cdot 6 + (5+4+3+2+1) = 15 + 42 + 15 = 72 \checkmark$

$$3.6 - y[n] = \sum_k x[k] g[n - 2k], \quad g[n] = u[n] - u[n - 4]$$

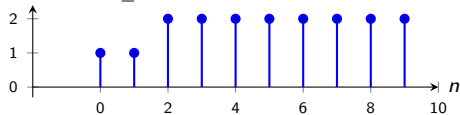
$g[n] = 1$ for $0 \leq n \leq 3$ (a length-4 pulse).

(a) $x[n] = \delta[n - 1]$: only $k = 1$ survives, so
 $y[n] = g[n - 2] = u[n - 2] - u[n - 6]$.



(b) $x[n] = u[n]$:

$$y[n] = \sum_{k \geq 0} g[n - 2k] = u[n] + u[n - 2].$$



3.7 – (a), (b): geometric * geometric

(a) $x = \alpha^n u[n]$, $h = \beta^n u[n]$, $\alpha \neq \beta$. Both are causal, so the sum runs over $0 \leq k \leq n$:

$$y[n] = \sum_{k=0}^n \alpha^k \beta^{n-k} = \beta^n \sum_{k=0}^n \left(\frac{\alpha}{\beta}\right)^k = \beta^n \frac{1 - (\alpha/\beta)^{n+1}}{1 - \alpha/\beta} = \boxed{\frac{\beta^{n+1} - \alpha^{n+1}}{\beta - \alpha} u[n]}$$

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(b) $x = h = \alpha^n u[n]$. Now every term is $\alpha^k \alpha^{n-k} = \alpha^n$, and there are $n + 1$ of them:

$$y[n] = \boxed{(n + 1) \alpha^n u[n]}$$

(Consistent with (a) as $\beta \rightarrow \alpha$: the difference quotient of $t \mapsto t^{n+1}$ is $(n + 1)\alpha^n$.)

3.7 – (c): $x[n] = (-\frac{1}{3})^n u[n-4]$, $h[n] = 4^n u[2-n]$

- $x[k] \neq 0$ iff $k \geq 4$. $h[n-k] = 4^{n-k} u[2-n+k] \neq 0$ iff $k \geq n-2$.
- So the sum starts at $K = \max(4, n-2)$ and runs to ∞ :

$$y[n] = \sum_{k=K}^{\infty} \left(-\frac{1}{3}\right)^k 4^{n-k} = 4^n \sum_{k=K}^{\infty} \left(-\frac{1}{12}\right)^k = 4^n \frac{(-1/12)^K}{1 + 1/12} = \frac{12}{13} 4^n \left(-\frac{1}{12}\right)^K$$

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- $n \leq 6$ ($K = 4$): $y[n] = \frac{12}{13} \frac{4^n}{12^4} = \frac{4^n}{13 \cdot 12^3} = \frac{4^n}{22464}$
- $n \geq 6$ ($K = n-2$): $y[n] = \frac{12}{13} 4^n \left(-\frac{1}{12}\right)^{n-2} = \frac{1728}{13} \left(-\frac{1}{3}\right)^n$

Both agree at $n = 6$: $4^6/22464 = \frac{1728}{13} \cdot \frac{1}{729} \approx 0.1823$ ✓